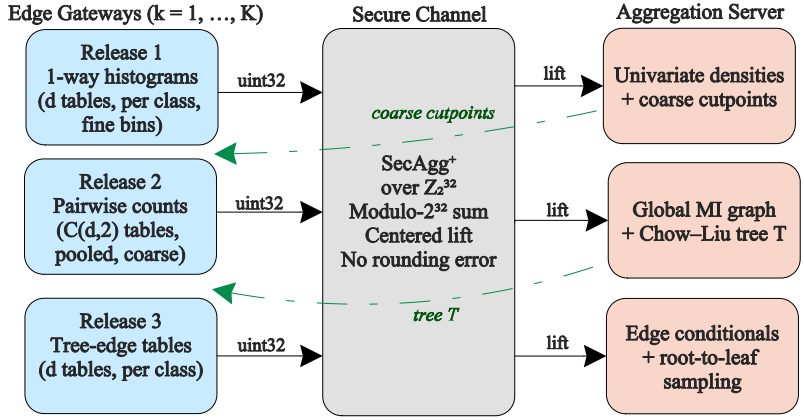

Edge Gateways (k = 1, …, K)
Secure Channel
Aggregation Server
Release 1
1-way histograms
(d tables, per class,
fine bins)
uint32
Release 2
Pairwise counts
(C(d,2) tables,
pooled, coarse)
uint32
Release 3
Tree-edge tables
(d tables, per class)
uint32
coarse cutpoints
SecAgg+
over Z2^32
Modulo-2^32 sum
Centered lift
No rounding error
tree T
lift
lift
lift
Univariate densities
+ coarse cutpoints
Global MI graph
+ Chow–Liu tree T
Edge conditionals
+ root-to-leaf
sampling